

**Theorem 3.7 (Algebra of Convergent Series).** Let  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} b_n$  be two convergent series with sums  $A$  and  $B$ , respectively. Then:

1. The series  $\sum_{n=1}^{\infty} (a_n + b_n)$  is convergent and

$$\sum_{n=1}^{\infty} (a_n + b_n) = A + B.$$

2. For every  $\lambda \in \mathbb{R}$ , the series  $\sum_{n=1}^{\infty} \lambda a_n$  is convergent and

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda A.$$

**Theorem 3.8 (Cauchy Criterion).** The series  $\sum_{n=1}^{\infty} a_n$  converges if and only if for every  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  such that

$$n, m \geq N \Rightarrow |s_n - s_m| < \epsilon,$$

where  $(s_n)$  denotes the sequence of partial sums. Thus, the series  $\sum_{n=1}^{\infty} a_n$  converges if and only if for every  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  such that

$$n > m \geq N \Rightarrow |a_{m+1} + a_{m+2} + \cdots + a_n| < \epsilon.$$

*Proof.* The series  $\sum_{n=1}^{\infty} a_n$  converges if and only if the sequence  $(s_n)$  of partial sums converges. By the Cauchy Criterion for sequences, this is equivalent to saying that  $(s_n)$  is a Cauchy sequence. Since

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n,$$

the second statement follows immediately.  $\square$

**Theorem 3.9.** Suppose that the series  $\sum_{n=1}^{\infty} a_n$  converges to  $s$ . Then, for every  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| < \epsilon.$$

*Proof.* Let  $\epsilon > 0$  be given. Since  $s_n \rightarrow s$ , there exists  $N \in \mathbb{N}$  such that  $|s - s_N| < \epsilon$ . Since  $\sum_{n=N+1}^{\infty} a_n = s - s_N$ , it follows that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| = |s - s_N| < \epsilon.$$

This completes the proof.  $\square$

**Theorem 3.10.** If the series  $\sum_{n=1}^{\infty} a_n$  converges, then  $\lim_{n \rightarrow \infty} a_n = 0$ .

*Proof.* Let  $s_n = \sum_{k=1}^n a_k$  be the sequence of partial sums. Since the series converges,  $(s_n)$  converges. Hence, by the Cauchy Criterion for sequences, for every  $\epsilon > 0$  there exists  $N \in \mathbb{N}$  such that

$$|s_n - s_m| < \epsilon, \quad \forall n, m \geq N.$$

Now let  $n \geq N + 1$ . Then  $n - 1 \geq N$ , and therefore

$$|a_n| = |s_n - s_{n-1}| < \epsilon.$$

Hence  $a_n \rightarrow 0$ . □

**Example 3.11.** Show that the series  $\sum_{n=1}^{\infty} a_n$  is divergent, where  $a_n = \frac{n}{n+1}$ .

*Proof.* Since  $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 \neq 0$ , the necessary condition for the convergence of a series is violated. Hence  $\sum_{n=1}^{\infty} a_n$  is divergent. □

**Theorem 3.12.** If the series  $\sum_{n=1}^{\infty} |a_n|$  converges, then the series  $\sum_{n=1}^{\infty} a_n$  also converges.

*Proof.* Let  $s_n = \sum_{k=1}^n a_k$  and  $\sigma_n = \sum_{k=1}^n |a_k|$  denote the sequences of partial sums of  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} |a_n|$ , respectively.

Since  $(\sigma_n)$  converges, it is a Cauchy sequence. Thus, for every  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  such that

$$|\sigma_n - \sigma_m| < \epsilon, \quad \forall n > m \geq N.$$

Then

$$|s_n - s_m| = |a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n| = \sigma_n - \sigma_m < \epsilon.$$

Hence,  $(s_n)$  is a Cauchy sequence. By the Cauchy Criterion for sequences,  $(s_n)$  converges.

Therefore,  $\sum_{n=1}^{\infty} a_n$  converges. □

**Example 3.13.** Show that the series  $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} a_n$ , where  $a_n = \frac{(-1)^{n+1}}{n}$ , is convergent.

*Proof.* Let  $s_n = a_1 + a_2 + \cdots + a_n$  be the sequence of partial sums. Then, for any  $p \in \mathbb{N}$ ,

$$|s_{n+p} - s_n| = \left| \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+3} - \cdots + (-1)^{p-1} \frac{1}{n+p} \right| < \frac{1}{n+1}.$$

Let  $\varepsilon > 0$ . Then  $|s_{n+p} - s_n| < \varepsilon$  whenever  $n > \frac{1}{\varepsilon} - 1$ . Choose  $m = \left\lfloor \frac{1}{\varepsilon} - 1 \right\rfloor + 2$ . Then  $m$  is a natural number and

$$|s_{n+p} - s_n| < \varepsilon$$

for all  $n \geq m$  and  $p = 1, 2, 3, \dots$ . Hence  $\{s_n\}$  is a Cauchy sequence. Since  $\mathbb{R}$  is complete,  $\{s_n\}$  converges. Therefore,  $\sum_{n=1}^{\infty} a_n$  is convergent. □

**Exercise 3.1.** 1. Consider the constant sequence  $a_n = c$ ,  $n \in \mathbb{N}$ , where  $c \in \mathbb{R}$  is fixed. Determine whether the series  $\sum_{n=1}^{\infty} a_n$  is convergent.

2. Find the sum of each of the following infinite series, if it exists.

$$(a) \sum_{n=1}^{\infty} \frac{1}{2^n + 2^{n+1}}.$$

$$(b) \sum_{n=1}^{\infty} \frac{3}{(3n-2)(3n+1)}.$$

3. Using the Cauchy criterion, determine whether each of the following series converges.

$$(a) \sum_{n=1}^{\infty} \frac{n}{2^n}.$$

$$(b) \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}.$$

### 3.2 Tests of convergence of a series of positive terms

The convergence or divergence of a series is determined by its sequence of partial sums. Since partial sums are often difficult to compute explicitly, we use *convergence tests*, which determine the nature of a series without evaluating its partial sums directly. We now study some of these tests.

**Definition 3.5.** A series  $\sum_{n=1}^{\infty} a_n$  is called a series of positive terms if  $a_n > 0$ , for all  $n \in \mathbb{N}$ .

**Theorem 3.14.** A series of positive real numbers  $\sum_{n=1}^{\infty} a_n$  is convergent if and only if the sequence of partial sums  $\{s_n\}$ ,  $s_n = \sum_{k=1}^n a_k$ , is bounded above.

**Remark 3.15.** Since  $s_n = a_1 + a_2 + \cdots + a_n$  is a monotone increasing sequence, if it is not bounded above, then  $\lim_{n \rightarrow \infty} s_n = \infty$ . Hence, the series  $\sum_{n=1}^{\infty} a_n$  diverges to  $\infty$ . Therefore, a series of positive real numbers either converges to a finite real number or diverges to  $\infty$ .

**Theorem 3.16 (Comparison Test).** Let  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} b_n$  be two series of nonnegative real numbers. Suppose that  $a_n \leq b_n$  for all  $n \in \mathbb{N}$ . Then:

1. If  $\sum_{n=1}^{\infty} b_n$  converges, then  $\sum_{n=1}^{\infty} a_n$  also converges.
2. If  $\sum_{n=1}^{\infty} a_n$  diverges, then  $\sum_{n=1}^{\infty} b_n$  also diverges.

**Example 3.17.** The series  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  is convergent. Since

$$0 \leq \frac{1}{n^3} \leq \frac{1}{n^2}, \quad \forall n \in \mathbb{N},$$

the Comparison Test implies that  $\sum_{n=1}^{\infty} \frac{1}{n^3}$  is also convergent.

**Example 3.18.** The harmonic series  $\sum_{n=1}^{\infty} \frac{1}{n}$  is divergent. Since

$$\frac{1}{n} \leq \frac{1}{\sqrt{n}}, \quad \forall n \in \mathbb{N},$$

the Comparison Test implies that  $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$  is also divergent.

**Theorem 3.19 (Limit Form of the Comparison Test).** Let  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} b_n$  be two series of positive real numbers. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l,$$

where  $0 < l < \infty$ , then the two series either both converge or both diverge.

**Example 3.20.** Test the convergence of the series

$$\frac{1+2}{2^3} + \frac{1+2+3}{3^3} + \frac{1+2+3+4}{4^3} + \cdots.$$

*Proof.* Let  $\sum_{n=1}^{\infty} a_n$  be the given series. Then

$$a_n = \frac{1+2+\cdots+(n+1)}{(n+1)^3} = \frac{(n+1)(n+2)}{2(n+1)^3} = \frac{n+2}{2(n+1)^2}.$$

Choose  $b_n = \frac{1}{n}$ . Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n(n+2)}{2(n+1)^2} = \frac{1}{2}.$$

Since  $\sum_{n=1}^{\infty} \frac{1}{n}$  diverges, the given series also diverges by the Limit Comparison Test. □

**Example 3.21.** Test the convergence of the series

$$\frac{1}{1 \cdot 2^2} + \frac{1}{2 \cdot 3^2} + \frac{1}{3 \cdot 4^2} + \cdots.$$

*Proof.* Let  $\sum_{n=1}^{\infty} a_n$  be the given series. Then  $a_n = \frac{1}{n(n+1)^2}$ . Choose  $b_n = \frac{1}{n^3}$ . Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} = 1.$$

Since  $\sum_{n=1}^{\infty} \frac{1}{n^3}$  converges, the given series also converges by the Limit Comparison Test. □

**Remark 3.22.** 1. If  $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ , then  $\sum_{n=1}^{\infty} a_n$  converges whenever  $\sum_{n=1}^{\infty} b_n$  converges.

2. If  $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ , then  $\sum_{n=1}^{\infty} a_n$  diverges whenever  $\sum_{n=1}^{\infty} b_n$  diverges.

**Theorem 3.23 ( $p$ -Series Test).** The series  $\sum_{n=1}^{\infty} \frac{1}{n^p}$  converges if  $p > 1$  and diverges if  $p \leq 1$ .

**Theorem 3.24 (d'Alembert's Ratio Test).** Let  $\sum_{n=1}^{\infty} a_n$  be a series of positive real numbers. Suppose that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = r.$$

Then:

1. If  $0 \leq r < 1$ , then the series  $\sum_{n=1}^{\infty} a_n$  converges.
2. If  $r > 1$ , then the series  $\sum_{n=1}^{\infty} a_n$  diverges.
3. If  $r = 1$ , then the test is inconclusive.

**Example 3.25.** 1. Consider the series  $\sum_{n=1}^{\infty} \frac{1}{n!}$ . Since

$$\frac{a_{n+1}}{a_n} = \frac{1}{n+1} \rightarrow 0,$$

the Ratio Test implies that the series converges.

2. Consider the series  $\sum_{n=1}^{\infty} \frac{2^n}{n}$ . Since

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{n+1} \cdot \frac{n}{2^n} = 2 \cdot \frac{n}{n+1} \rightarrow 2 > 1,$$

the Ratio Test implies that the series diverges.

3. Consider the harmonic series  $\sum_{n=1}^{\infty} \frac{1}{n}$ . Since

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \rightarrow 1,$$

the Ratio Test is inconclusive. Indeed, the harmonic series is divergent. On the other hand, for the series  $\sum_{n=1}^{\infty} \frac{1}{n^2}$ ,

$$\frac{a_{n+1}}{a_n} = \left( \frac{n}{n+1} \right)^2 \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

**Theorem 3.26 (Cauchy's Root Test).** Let  $\sum_{n=1}^{\infty} a_n$  be a series of positive real numbers. Suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = r.$$

Then:

1. If  $0 \leq r < 1$ , then the series  $\sum_{n=1}^{\infty} a_n$  converges.
2. If  $r > 1$ , then the series  $\sum_{n=1}^{\infty} a_n$  diverges.
3. If  $r = 1$ , then the test is inconclusive.

**Example 3.27.** 1. Consider the series  $\sum_{n=1}^{\infty} \frac{1}{2^n}$ . Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{2^n}} = \frac{1}{2},$$

the Root Test implies that the series converges.

2. Consider the series  $\sum_{n=1}^{\infty} 2^n$ . Since

$$\sqrt[n]{a_n} = \sqrt[n]{2^n} = 2,$$

the Root Test implies that the series diverges.

3. Consider the harmonic series  $\sum_{n=1}^{\infty} \frac{1}{n}$ . Since

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n}} = \frac{1}{n^{1/n}} \rightarrow 1,$$

the Root Test is inconclusive. Indeed, the harmonic series is divergent. On the other hand, for the series  $\sum_{n=1}^{\infty} \frac{1}{n^2}$ ,

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n^2}} = \frac{1}{n^{2/n}} \rightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

**Example 3.28.** Show that the series  $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$  converges.

*Proof.* Let  $a_n = \frac{2^n n!}{n^n}$ . Then

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = 2 \left( \frac{n}{n+1} \right)^n.$$

Since  $(1 + \frac{1}{n})^n \rightarrow e$ , we have

$$\left( \frac{n}{n+1} \right)^n = \frac{1}{(1 + \frac{1}{n})^n} \rightarrow \frac{1}{e}.$$

Hence

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{2}{e} < 1.$$

Therefore, by the Ratio Test, the series  $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$  converges. □

**Theorem 3.29 (Gauss's Test).** Let  $\sum_{n=1}^{\infty} a_n$  be a series of positive real numbers. Suppose  $\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + \frac{b_n}{n^p}$ , where  $p > 1$  and  $\{b_n\}$  is a bounded sequence. Then  $\sum_{n=1}^{\infty} a_n$  converges if  $\alpha > 1$  and diverges if  $\alpha \leq 1$ .

**Example 3.30.** Test the convergence of the series

$$1 + \frac{2^2}{3^2} + \frac{2^2 \cdot 4^2}{3^2 \cdot 5^2} + \frac{2^2 \cdot 4^2 \cdot 6^2}{3^2 \cdot 5^2 \cdot 7^2} + \cdots$$